

Outbreak Synth (OS)

Plan v3: what has been built and learned (Case Study 1, early outbreaks), and what comes next. OS studies past outbreaks, follows a new one day by day, says when it understands it, and only then releases synthetic patient data.

1. The idea in one paragraph

Something new appears, which we call **fob**. In Case Study 1 it is a new disease outbreak; in Case Study 2 (planned) it is a new vaccine. At first there are only a few records. OS combines them with what it learned from similar past outbreaks, keeps checking itself, and turns on a **Green Light** when its understanding is good enough. Only then does it generate synthetic patients. The contribution is the **method**: deciding what to learn from, how much to borrow and for how long, when it is ready, and explaining every decision.

2. Data used so far (all public, all cited in the repository)

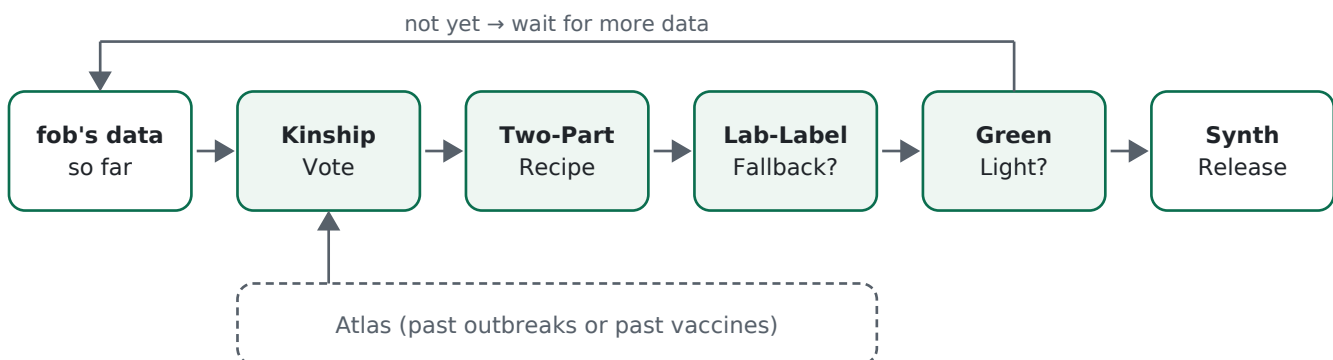
Source	What	Used for
Brazil SIVEP-Gripe (Ministry of Health, CC-BY)	24 outbreaks 2009–2022: H1N1 2009, flu and other-virus seasons, COVID-19 2020–22; about 2.15 million hospitalised patients	Outbreak Atlas; all replays
Mexico SISVER (Secretaría de Salud, free-use terms)	COVID-19 2020 and 2021: 326,886 and 297,685 hospitalised patients	Out-of-country test, settings fixed from Brazil
Kenema Ebola 2014 (Zenodo); H7N9 China 2013, MERS Korea 2015 (R package outbreaks, GPL)	83, 74 and 162 patients; age and sex only	Vote-only test on very different pathogens

3. The parts of OS, as built

Name	What it does now	Status
Outbreak Atlas	Past outbreaks mapped to shared fields (age, sex, race, state, 6 symptoms, 10 conditions). Fields a form never collected are 'missing'.	Built (Brazil + Mexico)
Kinship Vote	Each past outbreak predicts fob's new patients each week (patient mix + who dies, with fob's own death rate). Better predictions earn more votes. It compares only fields fob's form collects.	Built, tested
Stranger Flag	A 'none of the above' candidate learning from fob alone. Winning early means fob is new.	Built, tested
Lab-Label Fallback	New: if the lab says fob's pathogen group has no earlier member in the Atlas, OS uses an age-trend model until 20 deaths have been scored, then borrows.	Built, tested
Two-Part Recipe	Profile Model (who is hospitalised) + Severity Model (who dies). Severity: logistic regression with priors from voted relatives.	Built
Handover Rule	Borrowing strength $10 \times (1 - \text{Stranger share})$. Never borrowed: overall death rate, geography. A minimum restraint pulls towards 'no effect' (not towards relatives) so rare groups stay sensible.	Built, tuned on Brazil

Fresh-Days Check	Scores recipes on fob's newest unseen patients. As a recipe switcher it did not help (see section 5); kept as evidence in the Glass Box Report.	Built; no gain
Green Light	At least 20 deaths, and OS's risk ranking of fob's patients stable (rank correlation of 0.98 or more vs 8 days earlier) on two checks in a row, and not on the fallback.	Built, tested
Synth Release	After the Green Light: synthetic patients from the Two-Part Recipe. Symptoms and conditions come from a Dependency Chain (each given age band, sex and the fields before it). Every row is labelled synthetic with provenance.	Built, tested
Glass Box Report	An HTML page per outbreak and day: Green Light conditions, Kinship Vote, risk table (odds ratios, relatives' values, share still borrowed), recent evidence, synthetic vs real. Hindsight scores are kept separate.	Built
Time Machine Replay	Every outbreak replayed day by day as if new, hiding the future, to test all of the above.	Built

4. One step in the life of OS



Each day new fob data arrives. OS updates the Kinship Vote, applies the Lab-Label Fallback if needed, refits the Two-Part Recipe with the Handover Rule, and checks the Green Light. 'Not yet' means wait for more data.

5. What we learned (Time Machine Replays)

Question	Answer from the data
How long does real data alone take?	COVID-19 Brazil 2020: about 33 days to come within 0.02 AUC of the best possible. H1N1 2009: 56 days. In the first 3 weeks there is too little data to train at all.
Does the Kinship Vote find the right group?	Right group in 86% of 23 Brazil outbreaks in week 1 and 95% from week 4. When it is 95% sure, it is right 99% of the time.
Does the Stranger Flag catch a new disease?	COVID-19 flagged in week 3-4 in Brazil (749 patients) and week 4 in Mexico (1,262). It needs several hundred patients: it never fires in outbreaks under 200 patients.
Does borrowing help?	Familiar flu and other-virus seasons: at the plateau from day 0, while real data alone usually could not even train a model in 90 days. COVID-19 Brazil: plateau on day 28 instead of 34.

Is it safe for a new disease?	Without the Lab-Label Fallback, OS was worse than 'rank by age' in COVID's first 3 weeks. With it, never worse in Brazil. In Mexico it was below age-only for 34 of 90 days after the fallback ended, as was real data alone.
Is the Green Light honest?	Brazil: 22 of 23 outbreaks turned green within 90 days, 0 false (COVID-19 on day 44). Mexico: honest in both years (days 48 and 18).
Is the synthetic data useful?	Across 22 Brazil outbreaks a model trained on OS synthetic data scores AUC 0.795, against 0.730 for the real records OS had. For outbreaks with under 1,000 real records: 0.723 → 0.805 when added. CTGAN on the same records: 0.619, worse than real data in 22 of 22.
Is it realistic?	Synthetic patients match the real ones within about 0.4 points of death rate. With the Dependency Chain, links between symptoms and conditions are as close to later patients as the real data's own drift.

The key lesson: synthetic data from OS carries OS's understanding, no more and no less. It helps exactly when real data are scarce and borrowed knowledge is good. The Green Light decides when that is the case.

6. What did not work (kept in the record)

- Borrowing from all past outbreaks equally: it learned only 'older patients die more' (no better than ranking by age).
- Fresh-Days Check as a recipe switcher: it switched on a handful of patients and made early weeks worse. With a 100-death minimum it equals plain borrowing.
- 'Borrow only what relatives agree on' and 'borrow only the age pattern': both worse than plain borrowing.
- Outcome-based alarms for a new disease: there are almost no deaths in the first weeks, so they come too late.
- A misleading relative from the same group (H1N1 before the 2013 flu season) is still not caught early.

7. Open decisions

- **Equal start per group in the Kinship Vote.** Today each past outbreak starts equal, so groups with more members start ahead (flu 43%, COVID 20% before any data). With an equal start per group, MERS moves to 66% COVID-like. Adopting it means rerunning Steps 1-3.
- **When to leave the Lab-Label Fallback.** 20 deaths is a judgement call based on 3 new-pathogen outbreaks.
- **More truly new diseases.** The Atlas has about 4 distinct diseases and one new pathogen (COVID-19, in 2 countries). Trust claims rest on that.

8. Case Study 2: vaccine side-effect signals (planned, next)

Same method, different fob: a new vaccine rollout, week by week. Question: can OS confirm real side-effect signals earlier than standard safety statistics, with no more false alarms, and point to the groups at risk? Replay cases (dates to verify): myocarditis after mRNA vaccines; blood clots after viral-vector vaccines; narcolepsy after Pandemrix; intussusception after RotaShield. First step: check what public data can be downloaded (VAERS; Brazil's adverse-event reports), its licence and fields.

9. Limits we say up front

- Synthetic rows hold no more information than fob's real data plus what OS borrowed. Ten million rows is a format, not new knowledge.
- Synthetic data cannot reveal a side effect nobody has reported, and cannot replace clinical trials.

- Every synthetic file is labelled with its provenance; nothing is presented as real patients.
- The vote groups outbreaks by who gets sick and who dies, not by virus family (H7N9 influenza looks COVID-like). The lab label stays a separate input.
- Replays use final, cleaned datasets. Mexico is ordered by admission date, which ignores reporting delay.